MAE 5036: Advanced Computational Solid Mechanics II

Problem set 1
Due OCT 10, Saturday at 7:00 PM at Class

1. The elastic bar with the concentrated force in the domain (5PTs)

The **Dirac delta function** is also known as the unit impulse function, which satisfies the following property

$$\int_{-\infty}^{\infty} \delta(x - x_0) f(x)\, dx = f(x_0).$$

The delta function is named after physicist Paul Dirac, and has been applied routinely in physics and engineering to model point masses and concentrated loads.

The solution to the governing equation

$$\begin{cases} \dfrac{d}{dx}\left(EA\dfrac{du}{dx}\right) + P\delta(x - x_0) = 0, \quad x \in (0, 1), \\ u_{,x}(0) = 0, \\ u(1) = 0. \end{cases}$$

An analytical solution to this problem can be expressed using the Macaulay bracket, where P is the concentrated force applied at $x = x_0$.

(i) Show that the solution to the governing equation can be expressed as

$$u(x, x_0) = \frac{P}{EA}[(1 - x_0) - < x - x_0 >],$$

where

$$< x - x_0 >= \begin{cases} x - x_0, & x > x_0, \\ 0, & x \le x_0. \end{cases}$$

is the Macaulay bracket. The derivative of the Macaulay bracket is the Heaviside step function

$$H(x - x_0) = \begin{cases} 1, & x > x_0, \\ 0, & x \le x_0. \end{cases}$$

The derivative of the Heaviside function, in the distributional sense, is the Dirac delta:

$$\frac{d}{dx} H(x - x_0) = \delta(x - x_0).$$

(ii) Consider the following problem with the concentrated force $P = 100$ applied at $x = 0.5$, the distributed force is $b(x) = 2x$. Derive the analytical solution for the displacement field $u(x)$.

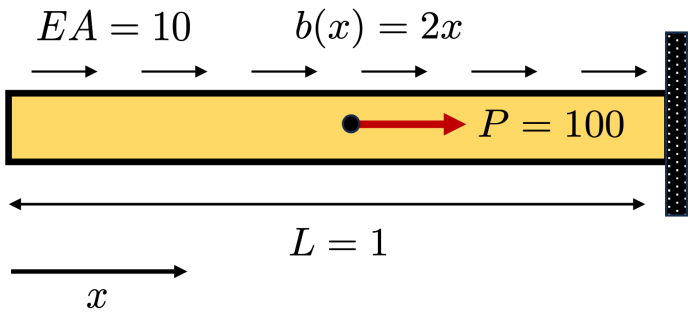


Figure 1: Problem 1

(iii) Derive the weak and Galerkin forms of the problem in (ii). Clearly define the trial and test spaces.

(iv) Use 2 linear finite elements with equal length to obtain the finite element solution $u^h(x)$ of (ii). Compare the finite element solution with the analytical solution derived in (ii).

Remark. *The solution is not sufficiently smooth to satisfy the classical strong form pointwise at $x = x_0$; however, it satisfies the governing equation in the weak sense.*

2. Bi-material elastic bar (5PTs)

The bi-material bar consists of two different materials joined together where

$$E(x) = \begin{cases} 100, & 0 \le x < 0.5, \\ 1, & 0.5 \le x \le 1, \end{cases}$$

and is subjected the distributed force $b(x) = x^2$.

It is intuitively expected that the displacement field will satisfy the following conditions:

$$\begin{cases} \dfrac{d}{dx}\left(E(x)A\dfrac{du}{dx}\right) + b(x) = 0, \quad x \in (0,1), \\ u(0) = 0, \\ u(1) = 0. \end{cases}$$

For simplicity, this question we assume the across sectional area is $A = 1$. However, due to the discontinuity in the material properties at $x = 0.5$, the displacement field may not be very smooth at this point.

(i) It is also natural to treat the problem as two subdomain problems, i.e. one for $0 \le x < 0.5$ and another for $0.5 \le x \le 1$. Then for each subdomain, the governing equation

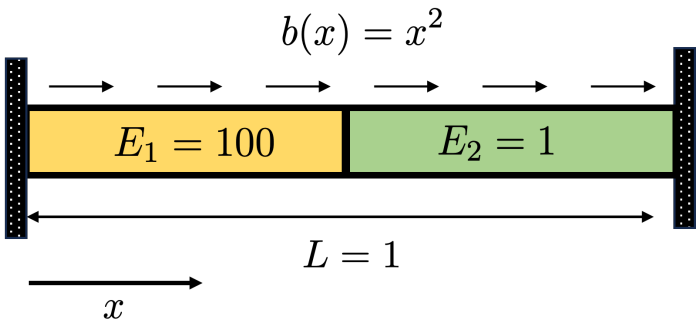


Figure 2: Problem 2

can be written as

$$\frac{d}{dx}\left(E(x)A\frac{du}{dx}\right) + b(x) = 0,$$

with the appropriate boundary and continuity conditions applied at $x = 0$, $x = 0.5$, and $x = 1$. The location $x = 0.5$ is where the material properties change, is called **the interface**. The jump condition at the interface requires that the displacement is continuous and the stress is continuous, i.e.

$$[u]_{0.5} = 0, \qquad [E(x)u_{,x}]_{0.5} = 0.$$

where

$$[f]_{0.5} = \lim_{x\to 0.5^+} f(x) - \lim_{x\to 0.5^-} f(x)$$

denotes the jump of a quantity f across the interface at $x = 0.5$.

Derive the analytical solution of this problem with the aid of jump condition.

(ii) Derive the weak form of this problem.

(iii) Derive the Galerkin weak form of this problem.

(iv) Consider using the linear finite element approximation for the displacement field (i.e. the Lagrangian linear shape function in terms of physical coordinates) and derive the corresponding element stiffness matrix and the force vector.

(v) Use 2 elements, assemble the global stiffness matrix and the global force vector **by hand**, find the global system of equations and solve for the numerical solution $u^h(x)$, compare it with the analytical solution derived in (i).

Remark. *This problem presents another example where the weak formulation is particularly useful due to the discontinuity in the material properties at the interface. The interface jump conditions are naturally incorporated into the weak formulation, allowing the solution to have a discontinuous strain while maintaining displacement and traction continuity across the interface.*

3. Numerical integration (5PTs)

A quadrature (the "fancy" name of integration) rule is an approximation of the definite integral of a function $f(x)$, usually stated as a weighted sum of function values at specified points within the domain of integration, say $\xi \in [-1, 1]$. In the derivation of integration points and weights for Gauss integration, the locations and weights are all variables which are obtained by maximizing the accuracy. The Gauss integration is the most useful approach to numerically evaluate the stiffness matrix and the force vector in finite element analysis. Consider the integration of the following function:

$$I = \int_a^b f(x)dx = \int_{-1}^{1} f\left(\frac{b-a}{2}\xi + \frac{a+b}{2}\right)\frac{dx}{d\xi}d\xi := \int_{-1}^{1} g(\xi)d\xi \approx \hat{\int}_{-1}^{1} g(\xi)d\xi = \sum_{I=1}^{GS} g(\xi_I)w_I.$$

where the change of variable from x to ξ is given by

$$x = \frac{b-a}{2}\xi + \frac{a+b}{2}$$

and $\hat{\int}$ is the numerical version of the integral, GS is the number of Gauss points, ξ_I are the Gauss points, and w_I are the corresponding weights.

(i) Suppose we want to use 1 point to generate the exact integral of a function, then there are two items to determine, the location of the Gauss point ξ_1 and the corresponding weight w_1. Then, these two unkowns can be found to exactly represent two types of integral:

$$g(\xi_1)w_1 = \int_{-1}^{1} 1d\xi, \quad \text{for } g(\xi) = 1,$$

$$g(\xi_1)w_1 = \int_{-1}^{1} \xi d\xi, \quad \text{for } g(\xi) = \xi$$

Solve the system of equations obtained in (i) to find the location of the Gauss point ξ_1 and the corresponding weight w_1.

(ii) Suppose we now have 2 Gauss points ξ_1 and ξ_2, think about what is the maximum degree of the polynomial that can be integrated exactly using these 2 points, find them out. (Solve them by hand)

(iii) Extend the idea to 3 Gauss points and determine the maximum degree of the polynomial that can be integrated exactly using these 3 points. (Solve them by hand, think about symmetricity)

(iv) Write a small computer program or do it by hand to assess the accuracy of Gauss integration of the following integral:

$$I = \int_0^1 \sin(\pi x)dx = \frac{2}{\pi}$$

Compare $\hat{\int}_0^1 \sin(\pi x)dx$ with the exact value $\frac{2}{\pi}$ by using 1, 2, and 3 Gauss points.

4. h refinement in finite element analysis (5PTs)

In this section, we will discuss the concepts of h refinement in finite element analysis. The h refinement refers to the process of reducing the element size to improve the accuracy of the solution, while the p refinement involves increasing the polynomial order of the shape functions within the elements. Both approaches aim to enhance the precision of the finite element solution, but they do so in different ways and have different computational implications. The source code is available at /src/FEM1D_linear_bar for this question. Also, please feel free to use or write your own computer program. The code template uses isoparametric approximation, i.e. we dont build the shape function in the physical coordinates. We instead build the shape functions in terms of the parametric coordinates, where the numerical integration is performed, in short:

$$x \approx x^h(\xi) = \sum_{i=1}^{n} N_i(\xi)x_i, \quad u(x) \approx u^h(x(\xi)) = \sum_{i=1}^{n} N_i(\xi)u_i,$$

where x_i and u_i are the nodal coordinates and nodal displacements, respectively.

(i) Read the code in the main.py and draw a flowchart of it.

(ii) Define the relative displacement error and energy:

$$e_u = \frac{\left(\int_0^L (u-u^h)^2 dx\right)^{1/2}}{\left(\int_0^L u^2 dx\right)^{1/2}},$$

$$e_E = \frac{\left(\int_0^L E(u_{,x}-u^h_{,x})^2 dx\right)^{1/2}}{\left(\int_0^L E(u_{,x})^2 dx\right)^{1/2}},$$

where u and $u_{,x}$ are the exact displacement and strain, and u^h and $u^h_{,x}$ are the numerical displacement and strain obtained from the finite element analysis, E is the Young's modulus.

Run the code with 2, 4, 6, 8, 10, 12, 14, 16 elements with uniform nodal space. Implement the calculation of e_u and e_E.

Plot the relative error against the meshsize in the log-log plot (i.e., plot log(e) against log(h)). Determine the slope of the graph.

Extra question. p refinement in finite element analysis (2PTs)

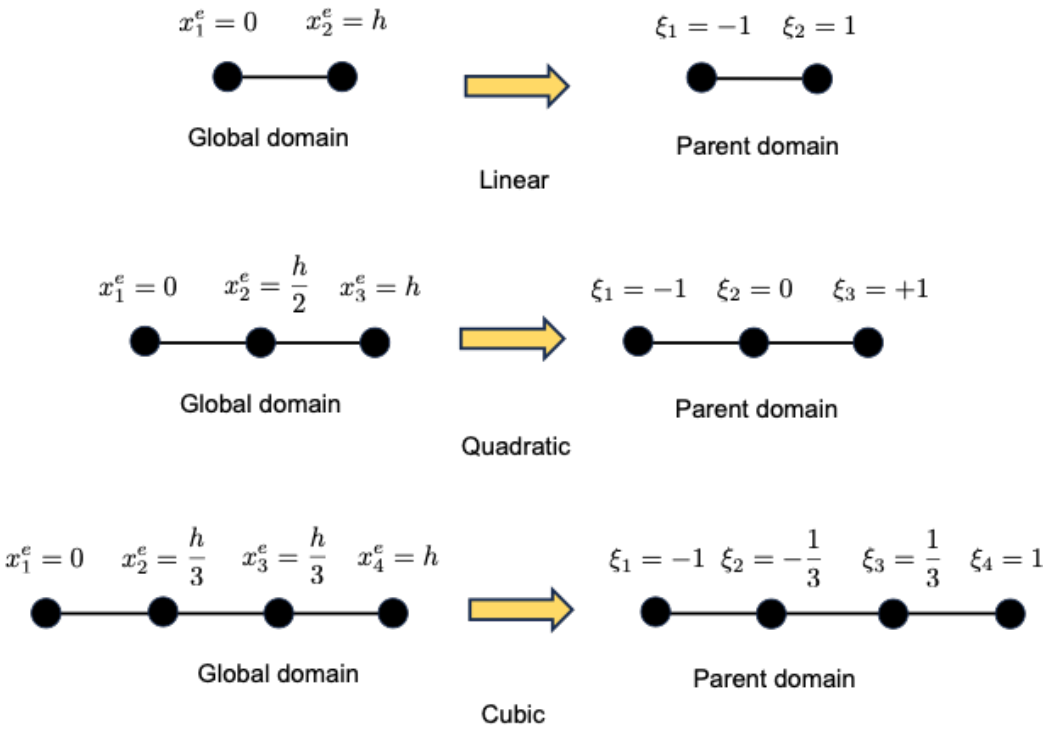


Figure 3: High order element

For more information of this question, read Hughes' book "The Finite Element Method: Linear Static and Dynamic Finite Element Analysis", page 126-129. In the class, we introduced the finite element approach with linear element, the linear element has two nodes per element, thus the interpolatory function is linear within each element. Subsequently, we can also upgrade the number of node in element by increasing the polynomial order of the shape functions, which is referred to as p refinement. The Lagrange polynomial shape function for multiple nodes within an element is given by:

$$N_i(\xi) = \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{\xi - \xi_j}{\xi_i - \xi_j}, \quad i = 1, 2, \ldots, n,$$

where n is the number of nodes in the element, ξ_i are the local coordinates of the nodes, and $N_i(\xi)$ are the shape functions associated with each node. You can easily verify the shape function here satisfies the Kronecker delta property: $N_i(\xi_j) = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

To understand the formulation better, the linear and quadratic shape functions can be explicitly written as:

- Linear element ($n = 2, \xi_1 = -1, \xi_2 = 1$):

$$N_1(\xi) = \frac{1-\xi}{2}, \quad N_2(\xi) = \frac{1+\xi}{2},$$

- Quadratic element ($n = 3, \xi_1 = -1, \xi_2 = 0, \xi_3 = 1$):

$$N_1(\xi) = \frac{\xi(\xi-1)}{2}, \quad N_2(\xi) = 1-\xi^2, \quad N_3(\xi) = \frac{\xi(\xi+1)}{2}.$$

(i) Upgrade the given code template or your own code, to implement p refinement by allowing for higher-order elements (e.g., quadratic elements), change the body force of the default problem to $\sin x$, bounadry conditions remain unchanged.

Redo (ii) in question 4 with quadratic, cubic, and quartic elements. Check the slopes for the error norm plot for each type of element. (HINT: in the linear 2 node element case, we have 2 points Gauss integration, however, for the higher-order elements, you may need more Gauss points to accurately integrate the element stiffness and force vector)